Technical Guide

Overview

The Ideathon Admin Platform is a Ruby on Rails application designed to support organizers in managing ProductTAMU Ideathons. The system provides tools for managing sponsors and partners, mentors and judges, FAQs and rules, along with activity tracking and notifications.

Tech Stack

Framework: Ruby on Rails (v8.0.4)
Language: Ruby
Database: PostgreSQL
Authentication: OmniAuth
Background Jobs: Active Job with Solid Queue
Email Delivery: Action Mailer with SMTP (SendGrid)
Containerization: Docker (single container)
Deployment: Heroku

System Architecture

This system follows the standard MVC (Model-View-Controller) architecture

- **Models:** Represent core entities (User, Ideathon, Sponsor, etc.)
- **Views:** Render UI pages for user interaction
- **Controllers:** Handle request/response logic

<u>Key Components:</u>

- **Authentication Layer:** OmniAuth-based login
- **User Interface:** CRUD operations for Ideathon data
- **Activity Logging System:** Tracks changes made by users
- **Email Notifications:** Sends updates via ActionMailer
- **Background Job Processing:** Handles async tasks (emails, imports)

Docker Setup

The application uses Docker to provide a consistent development and runtime environment. The application is built and run using a single Docker container.

Build the image

```
docker build -t 501-club-staging .
```

Run the container

```
docker run -p 3000:3000 501-club-staging
```

This will start the Rails server and expose it at http://localhost:3000

Environment Variables

Environment variables must be passed into the container at runtime

```
docker run -p 3000:3000 \
  -e DATABASE_URL=your_database_url \
  -e SENDGRID_API_KEY=your_sendgrid_key \
  501-club-staging
```

Database Setup

If the container does not automatically run migrations, execute:

```
docker run 501-club-staging rails db:create db:migrate
```

Stop the container

```
docker ps
docker stop <container_id>
```

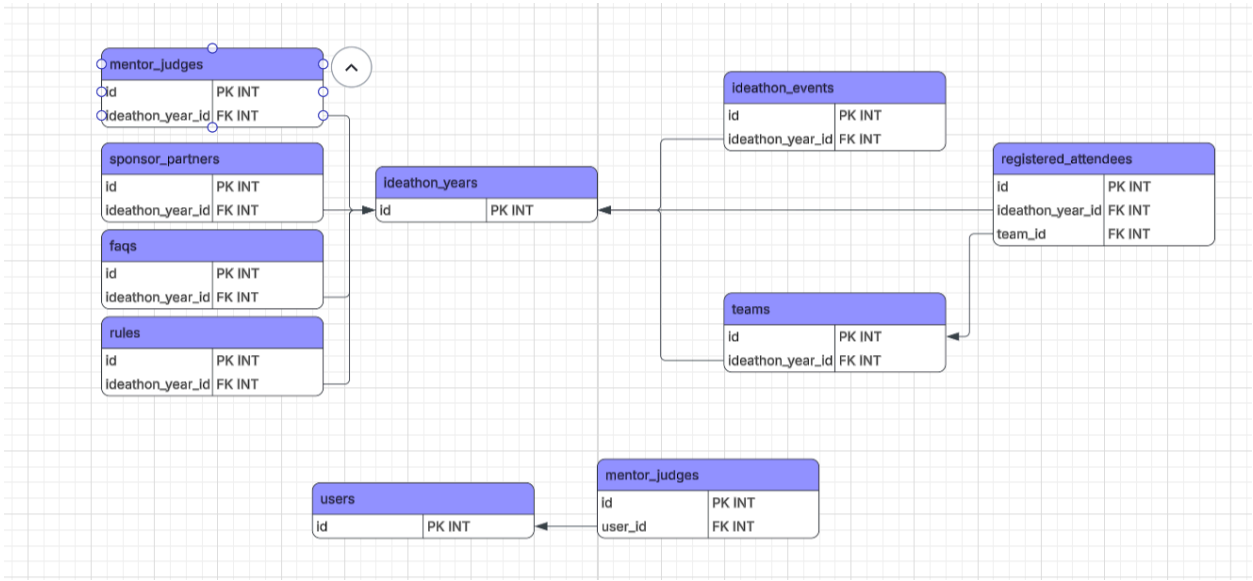
Database Design

Core Entities

- User: represents administrators and organizers
- Ideathon: represents an event instance
- Mentors_Judge: individuals assisting the event and//or participants
- Sponsors_Partner: organizations supporting the event
- FAQ: frequently asked questions
- Activity Log: records user actions for auditing

Relationships

- A User can create many activity logs
- An Ideathon can have many Sponsors, Mentors, and FAQs
- Activity logs are associated with users

Entity Relationship Diagram

Referential Integrity Diagram